

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 15C

CARBOXYLIC ACIDS (PHYSICAL PROPERTIES, PREPARATIONS & REACTIONS)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY: CARBONYLS, CARBOXYLIC ACIDS AND CHIRALITY
SUB-TOPIC:	15C: Carboxylic Acids (Physical Properties, Preparations & Reactions)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	228 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Carboxylic Acid Dimer Structure & Hydrogen Bonding****[4 Marks]****WCH14/01/Jan23/Q19**

Ethanoic acid, CH_3COOH , forms dimers in non-polar solvents and has an abnormally high boiling point (118°C).

- (a) Draw the structure of an ethanoic acid dimer showing all atoms, lone pairs, and hydrogen bonds. [3]

- (b) Explain why ethanoic acid has a higher boiling point than propan-1-ol (97°C) despite having lower molar mass. [1]

Q2. 2. Preparations of Carboxylic Acids: 3 Main Routes**[5 Marks]****WCH14/01/Oct22/Q20**

Propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$, can be prepared from alcohols, aldehydes, or nitriles.

- (a) State reagents and conditions for preparing propanoic acid from propan-1-ol. [2]

- (b) State reagents and conditions for preparing propanoic acid from propanenitrile, $\text{CH}_3\text{CH}_2\text{CN}$. [2]

- (c) Write a balanced equation for nitrile hydrolysis under acidic conditions. [1]

Q3. 3. Reactions of Carboxylic Acids as Weak Acids**[4 Marks]****WCH14/01/Jun22/Q21**

Ethanoic acid reacts with metals, alkalis, and carbonates.

- (a) Write a balanced equation for reaction of CH_3COOH with magnesium ribbon. [1]

(b) Write a balanced equation for reaction of CH_3COOH with sodium carbonate, Na_2CO_3 . [1]

(c) State observations for the sodium carbonate test and explain why it distinguishes carboxylic acids from phenols. [2]

Q4. 4. Reaction with Phosphorus(V) Chloride PCl_5

[4 Marks]

WCH14/01/Jan22/Q21

Carboxylic acids react with PCl_5 to form acyl chlorides.

(a) Write a balanced equation for the reaction of CH_3COOH with PCl_5 . [2]

(b) State observations and explain why this reaction MUST be carried out in a dry fume cupboard. [2]

Q5. 5. Esterification Reaction Mechanism & Acid Catalysis

[5 Marks]

WCH14/01/Oct21/Q20

Ethanoic acid reacts with ethanol in the presence of concentrated H_2SO_4 catalyst to form ethyl ethanoate.

(a) Write a balanced equation for esterification. [1]

(b) State the role of concentrated H_2SO_4 catalyst. [1]

(c) Isotopic labeling with oxygen-18 (^{18}O) in ethanol $\text{CH}_3\text{CH}_2\text{-}^{18}\text{OH}$ proves which C-O bond breaks. [3]

Q6. 6. Relative Acidity Comparison: Carboxylic Acids vs Phenols vs Alcohols

[5 Marks]
WCH14/01/Jun21/Q20

Compare pK_a values: Ethanoic acid (4.76), Phenol (9.95), Ethanol (16.0).

(a) Rank the three compounds in order of increasing acidity. [1]

(b) Explain the structural reasons for the differences in K_a . [4]

Q7. 7. Inductive Effects of Substituents on Carboxylic Acid Strength

[4 Marks]
WCH14/01/Jan21/Q20

Compare K_a values (mol dm^{-3}): Ethanoic acid (1.74×10^{-5}), Chloroethanoic acid (1.38×10^{-3}), Dichloroethanoic acid (5.13×10^{-2}), Trichloroethanoic acid (0.22).

(a) Explain why substituting chlorine atoms onto the α -carbon increases acid strength by 10,000-fold. [4]

Q8. 8. Reduction of Carboxylic Acids with LiAlH₄**[3 Marks]**
WCH14/01/Oct20/Q20

Propanoic acid is reduced to propan-1-ol using lithium tetrahydridoaluminate.

- (a) State reagents and conditions for reducing propanoic acid. [2]

- (b) Write a balanced equation using 4[H] for the reduction. [1]

Q9. 9. Physical Properties & Water Solubility Trend**[3 Marks]**
WCH14/01/Jun20/Q20

Methanoic, ethanoic, propanoic, and butanoic acids are completely soluble in water.

- (a) Explain why low molar mass carboxylic acids are highly soluble in water. [2]

- (b) Explain why hexadecanoic acid (palmitic acid, C₁₅H₃₁COOH) is insoluble in water. [1]

Q10. 10. Decarboxylation of Carboxylic Acid Salts**[4 Marks]**
WCH14/01/Jan20/Q20

Heating sodium ethanoate with soda lime (NaOH / CaO mixture) produces methane gas.

- (a) Write a balanced equation for soda lime decarboxylation of CH₃COONa. [2]

- (b) State the function of CaO in soda lime. [1]

- (c) State how decarboxylation alters carbon chain length. [1]

Q11. 11. Synthesis of Dicarboxylic Acids: Oxidation of Diols and Nitriles**[4 Marks]****WCH14/01/Oct19/Q20**

Hexanedioic acid, $\text{HOOC}(\text{CH}_2)_4\text{COOH}$ (adipic acid), is used to manufacture Nylon-6,6.

- (a) Write an equation for preparing hexanedioic acid by oxidation of hexane-1,6-diol with acidified $\text{K}_2\text{Cr}_2\text{O}_7$. [2]

- (b) Write an equation for preparing hexanedioic acid by hydrolysis of hexanedinitrile $\text{NC}(\text{CH}_2)_4\text{CN}$. [2]

Q12. 12. Reactions of Carboxylic Acids with Alkyl Hydroxides (Esterification Equilibrium)**[4 Marks]****WCH14/01/Jun19/Q21**

Calculating equilibrium yield of ester when 2.0 mol CH_3COOH and 1.0 mol $\text{C}_2\text{H}_5\text{OH}$ react ($K_c = 4.0$).

- (a) Set up the K_c expression in terms of x (moles of ester formed). [2]

- (b) Calculate x and percentage conversion of alcohol. [2]

Q13. 13. Infrared Spectroscopy Identification of Carboxylic Acids**[4 Marks]****WCH14/01/Jan19/Q22**

The IR spectrum of propanoic acid features two diagnostic absorption bands.

- (a) State the wavenumber range and appearance of the O–H stretching band in carboxylic acids. [2]

(b) State the wavenumber range of the C=O stretching band. [1]

(c) How does the carboxylic acid O-H band differ from an alcohol O-H band? [1]

Q14. 14. Mass Spectrometry Fragmentation of Carboxylic Acids

[4 Marks]

WCH14/01/sample/Q14

Propanoic acid $\text{CH}_3\text{CH}_2\text{COOH}$ ($M_r = 74$) undergoes characteristic fragmentation in mass spectrometry.

(a) Identify fragment ion peaks at m/z 45 and m/z 29. [4]

Q15. 15. Acidity of Substituted Benzoic Acids: Inductive vs Resonance Effects

[5 Marks]

WCH14/01/sample/Q15

pK_a values: Benzoic acid (4.20), 2-hydroxybenzoic acid / salicylic acid (2.97), 4-hydroxybenzoic acid (4.58).

(a) Explain why salicylic acid (2-hydroxybenzoic acid) is significantly more acidic than 4-hydroxybenzoic acid. [5]

Q16. 16. Grignard Reaction Synthesis of Carboxylic Acids using CO₂**[5 Marks]****WCH14/01/Sample/Q16**

Carboxylation of Grignard reagents: $R-MgBr + CO_2 \rightarrow R-COOMgBr \xrightarrow{(H^+/H_2O)} R-COOH$.

- (a) Write balanced equations for preparing benzoic acid from bromobenzene C₆H₅Br in 2 steps. [4]

- (b) State why solid CO₂ (dry ice) is used. [1]

Q17. 17. Acid-Catalysed Hydrolysis of Nitriles Mechanism**[5 Marks]****WCH14/01/Sample/Q17**

Hydrolysis of ethanenitrile CH₃CN to ethanoic acid CH₃COOH.

- (a) Write the intermediate amide formed during nitrile hydrolysis. [1]

- (b) Write two step-wise balanced equations for acid-catalysed nitrile hydrolysis. [4]

Q18. 18. Alkaline Hydrolysis of Nitriles followed by Acidification**[4 Marks]****WCH14/01/Sample/Q18**

Ethanenitrile CH₃CN heated under reflux with NaOH(aq), followed by adding dilute HCl.

- (a) Write equation for Step 1 (alkaline hydrolysis). [2]

(b) Write equation for Step 2 (acidification). [2]

Q19. 19. Test for Carboxylic Acid Functional Group: NaHCO₃ Effervescence

[3 Marks]

WCH14/01/Sample/Q19

Sodium hydrogen carbonate NaHCO₃ solution is added to an unknown liquid.

(a) State the observation and write the ionic equation for a positive test. [3]

Q20. 20. Formic Acid (Methanoic Acid) Unique Reducing Properties

[4 Marks]

WCH14/01/Sample/Q20

Methanoic acid, HCOOH, is unique among carboxylic acids because it contains both an aldehyde -CHO group and a carboxyl -COOH group.

(a) Explain why methanoic acid reduces Tollens' reagent and Fehling's solution, whereas ethanoic acid does not. [4]

Q21. 21. Oxidation of Methanoic Acid by Potassium Manganate(VII)

[4 Marks]

WCH14/01/Sample/Q21

Methanoic acid decolourises acidified KMnO_4 solution upon warming.

(a) Write a balanced ionic equation for the oxidation of HCOOH by MnO_4^- . [4]

Q22. 22. Oxidation of Ethanedioic Acid (Oxalic Acid) in Titrations

[4 Marks]

WCH14/01/Sample/Q22

Ethanedioic acid $(\text{COOH})_2$ is a primary standard used to standardize KMnO_4 .

(a) Write balanced ionic equation for $(\text{COOH})_2$ with MnO_4^- at 60°C . [4]

Q23. 23. Relative Boiling Points of Mono- vs Di-carboxylic Acids

[3 Marks]

WCH14/01/Sample/Q23

Boiling points: Propanoic acid (141°C), Propanedioic acid (decomposes at 135°C , high melting point 135°C).

(a) Explain why dicarboxylic acids have extremely high melting points and form polymer-like hydrogen-bonded networks. [3]

Q24. 24. Synthesis of Lactic Acid from Ethanol in 4 Steps

[5 Marks]

WCH14/01/Sample/Q24

Ethanol $\rightarrow$ Ethanal $\rightarrow$ 2-hydroxypropanenitrile $\rightarrow$ Lactic acid.

(a) Specify reagents and conditions for all 3 steps. [5]

Q25. 25. NMR Characteristics of Carboxylic Acid Protons**[3 Marks]****WCH14/01/Sample/Q25**

¹H NMR spectra of carboxylic acids feature a characteristic broad singlet.

- (a) State the chemical shift delta range for the -COOH proton in ¹H NMR. [1]

- (b) Explain why D₂O shake eliminates the -COOH proton peak. [2]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50

Q26. 26. A * Challenge: Acid Strength of Fluorinated vs Chlorinated Carboxylic Acids

[5 Marks]
WCH14/01/Hard/Q26

pKa values: CH₂FCOOH (2.59), CH₂ClCOOH (2.86), CH₂BrCOOH (2.90), CH₂ICOOH (3.18).

(a) Explain the trend in pK_a in terms of halogen electronegativity and electron-withdrawing inductive effect.
[5]

Q27. 27. A* Challenge: Distance Dependence of Inductive Effects on Acidity

[5 Marks]
WCH14/01/Hard/Q27

pKa values: 2-chlorobutanoic acid (2.86), 3-chlorobutanoic acid (4.05), 4-chlorobutanoic acid (4.52), Butanoic acid (4.82).

(a) Explain why moving the chlorine atom further from the -COOH group diminishes its acid-strengthening effect. [5]

Q28. 28. A* Challenge: Multi-Step Synthesis of Benzoic Acid from Methylbenzene

[5 Marks]
WCH14/01/Hard/Q28

Oxidation of alkylbenzenes with alkaline KMnO_4 .

(a) Write balanced equation for oxidation of methylbenzene $\text{C}_6\text{H}_5\text{CH}_3$ with alkaline KMnO_4 under reflux, followed by acidification. [4]

(b) Predict the oxidation product of ethylbenzene $C_6H_5CH_2CH_3$. [1]

Q29. 29. A* Challenge: Decarboxylation Mechanism of beta-Keto Acids

[5 Marks]
WCH14/01/Hard/Q29

beta-keto acids (e.g. 3-oxobutanoic acid, CH_3COCH_2COOH) undergo thermal decarboxylation readily at $50\text{ }^\circ\text{C}$ without soda lime.

(a) Draw the 6-membered cyclic transition state for decarboxylation of 3-oxobutanoic acid forming propan-2-ol enol. [5]

Q30. 30. A* Challenge: Retrosynthetic Design of Branched Carboxylic Acids

[5 Marks]
WCH14/01/Hard/Q30

Design two independent synthetic routes to prepare 2-methylbutanoic acid, $CH_3CH_2CH(CH_3)COOH$.

(a) Route 1: From 2-bromobutane via Grignard carboxylation with CO_2 . [2.5]

(b) Route 2: From 2-bromobutane via nitrile substitution and hydrolysis. [2.5]

Q31. 31. A* Challenge: Stereochemistry of Esterification of Chiral Alcohols

[5 Marks]
WCH14/01/Hard/Q31

Reaction of (R)-butan-2-ol with ethanoic acid in the presence of concentrated H₂SO₄.

(a) Predict whether the product sec-butyl ethanoate is optically active. [2]

(b) Explain why the configuration at the chiral C2 carbon atom is 100% retained. [3]

Q32. 32. A* Challenge: Acidic Hydrolysis Equilibrium Constant K_{hyd} for Esters

[5 Marks]
WCH14/01/Hard/Q32

Acidic hydrolysis: $\text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{CH}_3\text{OH}$ ($K_c = 0.25$).

(a) Explain why acid hydrolysis of esters never goes to 100% completion. [2]

(b) Calculate the equilibrium yield of CH_3COOH when 1.00 mol ester is refluxed with 5.00 mol water. [3]

**Q33. 33. A* Challenge: Alkaline Hydrolysis (Saponification)
Irreversibility**

[5 Marks]
WCH14/01/Hard/Q33

Alkaline hydrolysis: $\text{CH}_3\text{COOCH}_3 + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{CH}_3\text{OH}$.

(a) Explain why alkaline hydrolysis goes to 100% completion (irreversible) unlike acid hydrolysis. [5]

**Q34. 34. A* Challenge: Transesterification Reactions in
Biodiesel Production**

[5 Marks]
WCH14/01/Hard/Q34

Vegetable oil triglycerides (triesters of glycerol) are converted to biodiesel (fatty acid methyl esters, FAME).

(a) Write a general equation for transesterification of a triglyceride with methanol in the presence of NaOH catalyst. [3]

(b) Explain why biodiesel is considered carbon-neutral. [2]

Q35. 35. A* Challenge: Synthesis of Aspirin (Acetylsalicylic Acid)

[5 Marks]
WCH14/01/Hard/Q35

Aspirin is synthesized by reacting 2-hydroxybenzoic acid (salicylic acid) with ethanoic anhydride in the presence of H_3PO_4 .

(a) Write a balanced equation for aspirin synthesis. [2]

(b) Explain why ethanoic anhydride is preferred over ethanoyl chloride in industrial pharmaceutical manufacturing. [3]

Q36. 36. A* Challenge: Purification of Aspirin by Recrystallization & Melting Point QC

[5 Marks]
WCH14/01/Hard/Q36

Recrystallization of crude aspirin from ethanol/water solvent mixture.

(a) Describe the step-by-step procedure for recrystallization. [3]

(b) How does an impurity affect the melting point of aspirin (pure mp = 138–140 °C)? [2]

Q37. 37. A* Challenge: Quantitative Thin Layer Chromatography (TLC) of Aspirin**[5 Marks]**
WCH14/01/Hard/Q37

TLC monitoring of aspirin synthesis from salicylic acid.

- (a) Explain how R_f values are calculated: $R_f = \text{distance moved by spot} / \text{distance moved by solvent front}$. [2]

- (b) State how iron(III) chloride FeCl_3 solution is used to detect unreacted salicylic acid impurity. [3]

Q38. 38. A* Challenge: NMR Elucidation of Isomeric Esters $\text{C}_4\text{H}_8\text{O}_2$ **[5 Marks]**
WCH14/01/Hard/Q38

Four ester structural isomers have molecular formula $\text{C}_4\text{H}_8\text{O}_2$.

- (a) Draw and name the four ester isomers of $\text{C}_4\text{H}_8\text{O}_2$. [2]

- (b) Distinguish ethyl ethanoate $\text{CH}_3\text{COOCH}_2\text{CH}_3$ from methyl propanoate $\text{CH}_3\text{CH}_2\text{COOCH}_3$ using ^1H NMR splitting. [3]

Q39. 39. A* Challenge: Di-ester Hydrolysis Kinetics (Dimethyl Phthalate)**[5 Marks]**
WCH14/01/Hard/Q39

Alkaline hydrolysis of dimethyl phthalate (benzene-1,2-dicarboxylate dimethyl ester).

(a) Write balanced equation for complete saponification with NaOH. [3]

(b) Calculate moles of NaOH required to fully hydrolyse 0.050 mol of dimethyl phthalate. [2]

Q40. 40. A* Challenge: High Resolution ^{13}C NMR of Substituted Benzoic Acids**[5 Marks]**
WCH14/01/Hard/Q40

Predict the number of peaks in the ^{13}C NMR spectrum of:

(a) Benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$. [2.5]

(b) 4-methylbenzoic acid, $\text{CH}_3\text{-C}_6\text{H}_4\text{-COOH}$. [2.5]

Q41. 41. A* Challenge: Multi-Step Synthesis of Paracetamol from Phenol**[5 Marks]**
WCH14/01/Hard/Q41

Synthesis: Phenol $\rightarrow$ 4-nitrophenol $\rightarrow$ 4-aminophenol $\rightarrow$ Paracetamol (4-acetamidophenol).

(a) Step 1: Nitration of phenol with dilute HNO_3 . [1.5]

(b) Step 2: Reduction of 4-nitrophenol with Sn / concentrated HCl. [1.5]

(c) Step 3: Acetylation of 4-aminophenol with ethanoic anhydride. [2]

Q42. 42. A* Challenge: Mass Spectrometry Resolution of Carboxylic Acid Isomers**[5 Marks]**
WCH14/01/Hard/Q42

Distinguish pentanoic acid $\text{CH}_3(\text{CH}_2)_3\text{COOH}$ and 2,2-dimethylpropanoic acid $(\text{CH}_3)_3\text{CCOOH}$ using mass spec fragmentation.

(a) Identify base peak fragment in 2,2-dimethylpropanoic acid. [3]

(b) Identify fragment ion in pentanoic acid. [2]

Q43. 43. A* Challenge: Synthesis of Polyesters (PET / Terylene)**[5 Marks]**
WCH14/01/Hard/Q43

Condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

(a) Draw the repeat unit of Poly(ethylene terephthalate) (PET). [3]

(b) State why condensation polymerisation produces a small molecule byproduct (H_2O). [2]

Q44. 44. A* Challenge: Biodegradability of Polyesters vs Addition Polymers**[5 Marks]**
WCH14/01/Hard/Q44

Compare the environmental degradation of PET polyester vs Poly(ethene).

(a) Explain why PET is biodegradable via hydrolysis. [3]

(b) Explain why poly(ethene) is non-biodegradable. [2]

Q45. 45. A* Challenge: Synthesis of Acyl Chlorides using Thionyl Chloride SOCl₂

[5 Marks]
WCH14/01/Hard/Q45

Carboxylic acids react with thionyl chloride SOCl₂ to form acyl chlorides.

(a) Write balanced equation for CH₃COOH + SOCl₂. [2]

(b) Explain why SOCl₂ is preferred over PCl₅ in industrial acyl chloride preparations. [3]

Q46. 46. A* Challenge: Enthalpy of Combustion Trends in Monocarboxylic Acids

[5 Marks]
WCH14/01/Hard/Q46

Standard enthalpies of combustion Δ_c H° (kJ mol⁻¹): HCOOH (-255), CH₃COOH (-874), C₂H₅COOH (-1527), C₃H₇COOH (-2184).

(a) Calculate the average increment in Δ_c H° per -CH₂- methylene group. [2]

(b) Explain the linear relationship between Δ_c H° and carbon chain length. [3]

Q47. 47. A* Challenge: Synthesis of Poly(lactic acid) PLA via Cyclic Lactide**[5 Marks]**
WCH14/01/Hard/Q47

Lactic acid $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$ undergoes self-esterification.

- (a) Draw the structure of cyclic dimer (dilactide) formed by heating lactic acid. [3]

- (b) Describe ring-opening polymerisation of dilactide to high molar mass PLA. [2]

Q48. 48. A* Challenge: Gas Chromatography – Mass Spectrometry (GC–MS) of Fatty Acid Methyl Esters**[5 Marks]**
WCH14/01/Hard/Q48

FAME analysis in olive oil testing for adulteration.

- (a) Explain why non-volatile fatty acids must be converted to volatile methyl esters (FAME) before GC analysis. [2]

- (b) Explain how GC retention time t_R and MS fragmentation pattern identify oleic acid vs palmitic acid. [3]

Q49. 49. A* Challenge: Synthesis of Nylon-6,6 vs Terylene Comparison**[5 Marks]**
WCH14/01/Hard/Q49

Compare condensation polymerisation forming Nylon-6,6 polyamide vs Terylene polyester.

- (a) Identify monomers for Nylon-6,6 and draw its repeat unit. [3]

- (b) Compare intermolecular hydrogen bonding in Nylon-6,6 vs dipole-dipole forces in Terylene. [2]

Q50. 50. A* Challenge: Complete Carboxylic Acid Master Synthesis & Elucidation**[6 Marks]**
WCH14/01/Hard/Q50

An unknown organic solid A ($C_7H_6O_2$) is sparingly soluble in cold water but dissolves in $NaHCO_3(aq)$ with effervescence.

A reacts with PCl_5 forming B (C_7H_5ClO) with misty fumes.

A reacts with methanol in conc H_2SO_4 forming sweet-smelling ester C ($C_8H_8O_2$).

Nitration of A with conc HNO_3/H_2SO_4 yields single mononitro derivative D.

- (a) Deduce structures of A, B, C, D with full chemical justification. [6]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 15C

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Carboxylic Acid Dimer Structure & Hydrogen Bonding 1. (a) Two CH ₃ COOH molecules paired head-to-tail with TWO hydrogen bonds between O–H of one and C=O of the other (3). 1. (b) Dimers effectively double the molecular mass in liquid phase and form 2 hydrogen bonds per dimer pair (1).	[4]
Q2	Q2: 2. Preparations of Carboxylic Acids: 3 Main Routes 2. (a) Excess K ₂ Cr ₂ O ₇ and concentrated H ₂ SO ₄ , heat under reflux (2). 2. (b) Dilute HCl(aq) or H ₂ SO ₄ (aq), heat under reflux (2). 2. (c) CH ₃ CH ₂ CN + 2H ₂ O + HCl → CH ₃ CH ₂ COOH + NH ₄ Cl (1).	[5]
Q3	Q3: 3. Reactions of Carboxylic Acids as Weak Acids 3. (a) 2CH ₃ COOH + Mg → (CH ₃ COO) ₂ Mg + H ₂ (1). 3. (b) 2CH ₃ COOH + Na ₂ CO ₃ → 2CH ₃ COONa + H ₂ O + CO ₂ (1). 3. (c) Effervescence / gas bubbles of CO ₂ turning limewater cloudy (1). Carboxylic acids are strong enough acids to decompose Na ₂ CO ₃ , whereas weaker phenols (pK _a 10) do not react (1).	[4]
Q4	Q4: 4. Reaction with Phosphorus(V) Chloride PCl ₅ 4. (a) CH ₃ COOH + PCl ₅ → CH ₃ COCl + POCl ₃ + HCl(g) (2). 4. (b) Vigorous reaction with misty white fumes of HCl(g) (1). PCl ₅ and acyl chlorides react violently with moisture in air (1).	[4]
Q5	Q5: 5. Esterification Reaction Mechanism & Acid Catalysis 5. (a) CH ₃ COOH + CH ₃ CH ₂ OH ⇌ CH ₃ COOCH ₂ CH ₃ + H ₂ O (1). 5. (b) Catalyst (protonates C=O) and dehydrating agent to absorb H ₂ O and shift equilibrium right (1). 5. (c) The I ⁸ O label ends up ENTIRELY in the ester product (CH ₃ CO–I ⁸ OCH ₂ CH ₃), proving the C–OH bond of the CARBOXYLIC ACID breaks, NOT the alcohol O–H bond (3).	[5]
Q6	Q6: 6. Relative Acidity Comparison: Carboxylic Acids vs Phenols vs Alcohols 6. (a) Ethanol < Phenol < Ethanoic acid (1). 6. (b) Ethanoate anion CH ₃ COO [–] has negative charge delocalized equally over TWO electronegative oxygen atoms via resonance (2). Phenoxide anion C ₆ H ₅ O [–] has negative charge delocalized into benzene pi ring (1). Ethoxide anion C ₂ H ₅ O [–] has negative charge localized on single oxygen with electron-donating alkyl group (+I effect) destabilising anion (1).	[5]
Q7	Q7: 7. Inductive Effects of Substituents on Carboxylic Acid Strength 7. (a) Chlorine is highly electronegative and exerts an electron-withdrawing inductive effect (–I effect) (1). Withdraws electron density along C–C and C–O bonds away from carboxylate group (1). Disperses and stabilizes the negative charge on –COO [–] anion (1). Shifts equilibrium RCOOH ⇌ RCOO [–] + H ⁺ right, increasing K _a (1).	[4]
Q8	Q8: 8. Reduction of Carboxylic Acids with LiAlH ₄ 8. (a) LiAlH ₄ in dry ether solvent (1), followed by dilute acid hydrolysis (1). 8. (b) CH ₃ CH ₂ COOH + 4[H] → CH ₃ CH ₂ CH ₂ OH + H ₂ O (1).	[3]
Q9	Q9: 9. Physical Properties & Water Solubility Trend 9. (a) Carboxylic acid group –COOH forms multiple strong hydrogen bonds with water molecules (both C=O and –OH act as H-bond acceptors/donors) (2). 9. (b) Long non-polar hydrocarbon tail (C ₁₅ H ₃₁ –) dominates, disrupting water hydrogen bonding network (1).	[3]
Q10	Q10: 10. Decarboxylation of Carboxylic Acid Salts 10. (a) CH ₃ COONa + NaOH → CH ₄ (g) + Na ₂ CO ₃ (s) (2). 10. (b) CaO prevents NaOH from melting rapidly and attacking reaction glassware (1). 10. (c) Shortens carbon chain by ONE carbon atom (decarbonylation) (1).	[4]

Q11	Q11: 11. Synthesis of Dicarboxylic Acids: Oxidation of Diols and Nitriles 11. (a) $\text{HO}(\text{CH}_2)_6\text{OH} + 4[\text{O}] \rightarrow \text{HOOC}(\text{CH}_2)_4\text{COOH} + 2\text{H}_2\text{O}$ (2). 11. (b) $\text{NC}(\text{CH}_2)_4\text{CN} + 4\text{H}_2\text{O} + 2\text{HCl} \rightarrow \text{HOOC}(\text{CH}_2)_4\text{COOH} + 2\text{NH}_4\text{Cl}$ (2).	[4]
Q12	Q12: 12. Reactions of Carboxylic Acids with Alkyl Hydroxides (Esterification Equilibrium) 12. (a) $K_c = x^2 / ((2.0-x)(1.0-x)) = 4.0 \Rightarrow x^2 = 4.0(2 - 3x + x^2) \Rightarrow 3x^2 - 12x + 8 = 0$ (2). 12. (b) $x = (12 - \sqrt{144 - 96}) / 6 = (12 - 6.928) / 6 = 0.845 \text{ mol}$ (1). % conversion = $(0.845 / 1.0) \times 100\% = 84.5\%$ (1).	[4]
Q13	Q13: 13. Infrared Spectroscopy Identification of Carboxylic Acids 13. (a) Very broad absorption at $2500\text{--}3300 \text{ cm}^{-1}$ overlapping C-H stretching region (2). 13. (b) Strong, sharp C=O peak at $1700\text{--}1725 \text{ cm}^{-1}$ (1). 13. (c) Alcohol O-H is a smooth broad peak at $3200\text{--}3600 \text{ cm}^{-1}$ without overlapping C-H region (1).	[4]
Q14	Q14: 14. Mass Spectrometry Fragmentation of Carboxylic Acids 14. (a) m/z 45 = $[\text{COOH}]^+$ carboxyl fragment ion (loss of C_2H_5 , 29 Da) (2). m/z 29 = $[\text{C}_2\text{H}_5]^+$ ethyl carbocation fragment ion (loss of COOH , 45 Da) (2).	[4]
Q15	Q15: 15. Acidity of Substituted Benzoic Acids: Inductive vs Resonance Effects 15. (a) Salicylic acid ($-\text{OH}$ at ortho position) forms an INTRAMOLECULAR HYDROGEN BOND between ortho $-\text{OH}$ group and carboxylate $-\text{COO}^-$ anion (2). Intramolecular H-bonding stabilizes the conjugate base anion immensely (2). In 4-hydroxybenzoic acid (para position), intramolecular H-bonding is geometrically impossible, and $-\text{OH}$ donates electrons by resonance (+M), weakening acidity (1).	[5]
Q16	Q16: 16. Grignard Reaction Synthesis of Carboxylic Acids using CO_2 16. (a) Step 1: $\text{C}_6\text{H}_5\text{Br} + \text{Mg} \xrightarrow{\text{dry ether}} \text{C}_6\text{H}_5\text{MgBr}$ (2). Step 2: $\text{C}_6\text{H}_5\text{MgBr} + \text{CO}_2 \rightarrow \text{C}_6\text{H}_5\text{COOMgBr} \xrightarrow{\text{dilute HCl}} \text{C}_6\text{H}_5\text{COOH} + \text{MgBrCl}$ (2). 16. (b) Dry ice provides cold temperature (-78°C) and anhydrous reactant source (1).	[5]
Q17	Q17: 17. Acid-Catalysed Hydrolysis of Nitriles Mechanism 17. (a) Ethanamide, CH_3CONH_2 (1). 17. (b) Step 1: $\text{CH}_3\text{CN} + \text{H}_2\text{O} + \text{H}^+ \rightarrow \text{CH}_3\text{CONH}_2 + \text{H}^+$ (amide intermediate) (2). Step 2: $\text{CH}_3\text{CONH}_2 + \text{H}_2\text{O} + \text{H}^+ \rightarrow \text{CH}_3\text{COOH} + \text{NH}_4^+$ (carboxylic acid + ammonium ion) (2).	[5]
Q18	Q18: 18. Alkaline Hydrolysis of Nitriles followed by Acidification 18. (a) $\text{CH}_3\text{CN} + \text{NaOH} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COONa} + \text{NH}_3(\text{g})$ (2). 18. (b) $\text{CH}_3\text{COONa} + \text{HCl} \rightarrow \text{CH}_3\text{COOH} + \text{NaCl}$ (2).	[4]
Q19	Q19: 19. Test for Carboxylic Acid Functional Group: NaHCO_3 Effervescence 19. (a) Vigorous effervescence / gas bubbles turning limewater milky (1). Ionic equation: $\text{RCOOH}(\text{aq}) + \text{HCO}_3^-(\text{aq}) \rightarrow \text{RCOO}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ (2).	[3]
Q20	Q20: 20. Formic Acid (Methanoic Acid) Unique Reducing Properties 20. (a) HCOOH structure $\text{H}-\text{C}(=\text{O})-\text{OH}$ contains an aldehyde C-H group attached to $\text{C}=\text{O}$ (2). It is easily oxidized further to carbonic acid / CO_2 and H_2O : $\text{HCOOH} + [\text{O}] \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ (1). CH_3COOH lacks an aldehyde proton and cannot be oxidized by Tollens' or Fehling's (1).	[4]
Q21	Q21: 21. Oxidation of Methanoic Acid by Potassium Manganate(VII) 21. (a) $5\text{HCOOH} + 2\text{MnO}_4^- + 6\text{H}^+ \rightarrow 5\text{CO}_2 + 2\text{Mn}^{2+} + 8\text{H}_2\text{O}$ (4). Purple MnO_4^- decolourises to colourless Mn^{2+} .	[4]
Q22	Q22: 22. Oxidation of Ethanedioic Acid (Oxalic Acid) in Titrations 22. (a) $5(\text{COOH})_2 + 2\text{MnO}_4^- + 6\text{H}^+ \rightarrow 10\text{CO}_2 + 2\text{Mn}^{2+} + 8\text{H}_2\text{O}$ (4).	[4]
Q23	Q23: 23. Relative Boiling Points of Mono- vs Di-carboxylic Acids 23. (a) Dicarboxylic acids have TWO $-\text{COOH}$ groups per molecule (1), forming extensive 3D intermolecular hydrogen-bonded networks in crystal lattice (2).	[3]
Q24	Q24: 24. Synthesis of Lactic Acid from Ethanol in 4 Steps 24. (a) Step 1: $\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}_2\text{SO}_4$, distill off ethanal (1). Step 2: $\text{HCN} + \text{KCN}$, pH 8 $\rightarrow$ 2-hydroxypropanenitrile (2). Step 3: Dilute HCl , reflux $\rightarrow$ 2-hydroxypropanoic acid (lactic acid) (2).	[5]

Q25	Q25: 25. NMR Characteristics of Carboxylic Acid Protons 25. (a) δ 10.0 – 12.0 ppm (1). 25. (b) Labile -COOH proton rapidly exchanges with deuterium in D_2O : $\text{RCOOH} + \text{D}_2\text{O} \rightleftharpoons \text{RCOOD} + \text{HOD}$. Deuterium does not absorb in ^1H NMR region, causing peak to disappear (2).	[3]
Q26	Q26: 26. A* Challenge: Acid Strength of Fluorinated vs Chlorinated Carboxylic Acids 26. (a) Electronegativity decreases down halogen group: $\text{F} (4.0) > \text{Cl} (3.0) > \text{Br} (2.8) > \text{I} (2.5)$ (1). Fluorine exerts strongest $-I$ electron-withdrawing inductive effect (1). Pulls electron density along C-C and C-O bonds most effectively (1). Stabilizes carboxylate anion -COO- to greatest extent (1). Shifts equilibrium right, giving highest K_a and lowest $\text{p}K_a$ (strongest acid) (1).	[5]
Q27	Q27: 27. A* Challenge: Distance Dependence of Inductive Effects on Acidity 27. (a) Electron-withdrawing inductive effect ($-I$) operates through sigma bonds (1). Inductive effect diminishes rapidly with increasing distance / number of intervening sigma bonds (2). In 2-chlorobutanoic acid, Cl is on C_2 (adjacent to -COOH), exerting maximum charge stabilization on -COO- (1). In 4-chlorobutanoic acid, Cl is 3 carbon atoms away, so $-I$ effect on -COO- is negligible (1).	[5]
Q28	Q28: 28. A* Challenge: Multi-Step Synthesis of Benzoic Acid from Methylbenzene 28. (a) Step 1: $\text{C}_6\text{H}_5\text{CH}_3 + 2\text{KMnO}_4 \rightarrow \text{C}_6\text{H}_5\text{COOK} + 2\text{MnO}_2(\text{s}) + \text{KOH} + \text{H}_2\text{O}$ (2). Step 2: $\text{C}_6\text{H}_5\text{COOK} + \text{HCl} \rightarrow \text{C}_6\text{H}_5\text{COOH}(\text{s}) + \text{KCl}$ (2). 28. (b) Benzoic acid $\text{C}_6\text{H}_5\text{COOH}$ (side chain cleaved regardless of length if $\alpha\text{-H}$ present) (1).	[5]
Q29	Q29: 29. A* Challenge: Decarboxylation Mechanism of beta-Keto Acids 29. (a) Intramolecular hydrogen bond between C=O oxygen of beta-keto group and -COOH proton forms 6-membered ring (2). Concerted cyclic 6-electron rearrangement (1): CO_2 is eliminated (1), forming enol intermediate $\text{CH}_2=\text{C}(\text{OH})\text{CH}_3$ which tautomerizes to propanone (1).	[5]
Q30	Q30: 30. A* Challenge: Retrosynthetic Design of Branched Carboxylic Acids 30. (a) $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{Br} + \text{Mg} \rightarrow \text{Grignard} \rightarrow \text{--}(\text{CO}_2)\text{--} \rightarrow \text{--}(\text{H}+)\text{--} \rightarrow \text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{COOH}$ (2.5). 30. (b) $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{Br} + \text{KCN} \rightarrow \text{--}(\text{ethanolic})\text{--} \rightarrow \text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CN} \rightarrow \text{--}(\text{H}+/\text{H}_2\text{O}, \text{reflux})\text{--} \rightarrow \text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{COOH}$ (2.5).	[5]
Q31	Q31: 31. A* Challenge: Stereochemistry of Esterification of Chiral Alcohols 31. (a) Optically active (100% (R) -configuration retained) (2). 31. (b) Esterification mechanism involves cleavage of acyl C-OH bond of ethanoic acid (1). The C-O bond attached to the chiral C_2 carbon of (R) -butan-2-ol is NEVER broken during reaction (2).	[5]
Q32	Q32: 32. A* Challenge: Acidic Hydrolysis Equilibrium Constant K_{hyd} for Esters 32. (a) Reaction is reversible (equilibrium established with $K_c = 0.25$) (2). 32. (b) $K_c = \frac{x^2}{(1-x)(5-x)} = 0.25 \Rightarrow x^2 = 0.25(5 - 6x + x^2) \Rightarrow 0.75x^2 + 1.5x - 1.25 = 0 \Rightarrow 3x^2 + 6x - 5 = 0 \Rightarrow x = \frac{-6 \pm \sqrt{36 + 60}}{6} = \frac{-6 + 9.798}{6} = 0.633 \text{ mol}$ (63.3% yield) (3).	[5]
Q33	Q33: 33. A* Challenge: Alkaline Hydrolysis (Saponification) Irreversibility 33. (a) Alkaline hydrolysis produces ethanoate ions $\text{CH}_3\text{COO-}$ (1). Ethanoate ion is negatively charged and resonance-stabilized (1). $\text{CH}_3\text{COO-}$ cannot undergo nucleophilic attack by alcohol CH_3OH (electrostatic repulsion / non-electrophilic) (2). Removal of CH_3COOH as $\text{CH}_3\text{COO-}$ shifts equilibrium 100% to the right (irreversible) (1).	[5]
Q34	Q34: 34. A* Challenge: Transesterification Reactions in Biodiesel Production 34. (a) Triglyceride + $3\text{CH}_3\text{OH} \rightarrow \text{--}(\text{NaOH})\text{--} \rightarrow \text{Glycerol} + 3 \text{ Fatty Acid Methyl Esters (FAME)}$ (3). 34. (b) CO_2 released during combustion of biodiesel was recently absorbed from the atmosphere by oil crops during photosynthesis (2).	[5]
Q35	Q35: 35. A* Challenge: Synthesis of Aspirin (Acetylsalicylic Acid) 35. (a) $\text{C}_6\text{H}_4(\text{OH})\text{COOH} + (\text{CH}_3\text{CO})_2\text{O} \rightarrow \text{C}_6\text{H}_4(\text{OCOCH}_3)\text{COOH} + \text{CH}_3\text{COOH}$ (2). 35. (b) Ethanoic anhydride is cheaper, less corrosive, less vulnerable to hydrolysis, and does not produce toxic, hazardous corrosive $\text{HCl}(\text{g})$ fumes (3).	[5]

Q36	Q36: 36. A* Challenge: Purification of Aspirin by Recrystallization & Melting Point QC 36. (a) Dissolve crude aspirin in minimum volume of hot solvent (1). Hot filter to remove insoluble impurities (1). Cool filtrate in ice bath to crystallize pure aspirin, suction filter, and dry (1). 36. (b) Impurities lower the melting point and broaden the melting point range (e.g. 130–135 °C) (2).	[5]
Q37	Q37: 37. A* Challenge: Quantitative Thin Layer Chromatography (TLC) of Aspirin 37. (a) R_f = distance moved by solute spot / distance moved by solvent front (2). 37. (b) $FeCl_3$ reacts with phenol -OH group in salicylic acid to form a purple complex (2). Pure aspirin lacks a phenol -OH group and gives NO colour change with $FeCl_3$ (1).	[5]
Q38	Q38: 38. A* Challenge: NMR Elucidation of Isomeric Esters $C_4H_8O_2$ 38. (a) Ethyl ethanoate, methyl propanoate, propyl formate, isopropyl formate (2). 38. (b) Ethyl ethanoate: singlet at 2.0 ppm (CH_3CO), quartet at 4.1 ppm ($-OCH_2-$), triplet at 1.2 ppm ($-CH_3$) (1.5). Methyl propanoate: singlet at 3.7 ppm ($-OCH_3$), quartet at 2.3 ppm ($-CH_2CO-$), triplet at 1.1 ppm ($-CH_3$) (1.5).	[5]
Q39	Q39: 39. A* Challenge: Di-ester Hydrolysis Kinetics (Dimethyl Phthalate) 39. (a) $C_6H_4(COOCH_3)_2 + 2NaOH \rightarrow C_6H_4(COO-Na^+)_2 + 2CH_3OH$ (3). 39. (b) Moles $NaOH = 2 \times 0.050 = 0.100$ mol (2).	[5]
Q40	Q40: 40. A* Challenge: High Resolution ^{13}C NMR of Substituted Benzoic Acids 40. (a) Benzoic acid has 5 non-equivalent carbon environments (ipso C_1 , ortho C_2/C_6 , meta C_3/C_5 , para C_4 , carboxyl $C=O$) $\Rightarrow$ 5 peaks (2.5). 40. (b) 4-methylbenzoic acid has 5 non-equivalent carbon environments (methyl C , ipso C_4 , meta C_3/C_5 , ortho C_2/C_6 , ipso C_1 , carboxyl $C=O$) $\Rightarrow$ 5 peaks (2.5).	[5]
Q41	Q41: 41. A* Challenge: Multi-Step Synthesis of Paracetamol from Phenol 41. (a) Dilute HNO_3 at room temp forms 4-nitrophenol (1.5). 41. (b) Sn and conc HCl , heat under reflux, followed by $NaOH$ (1.5). 41. (c) 4-aminophenol + $(CH_3CO)_2O \rightarrow$ 4-acetamidophenol (Paracetamol) + CH_3COOH (2).	[5]
Q42	Q42: 42. A* Challenge: Mass Spectrometry Resolution of Carboxylic Acid Isomers 42. (a) 2,2-dimethylpropanoic acid cleaves alpha to $C=O$, losing $COOH$ (45 Da) to give massive stable tert-butyl carbocation $[(CH_3)_3C]^+$ base peak at m/z 57 (3). 42. (b) Pentanoic acid undergoes McLafferty rearrangement yielding fragment ion at m/z 60 (2).	[5]
Q43	Q43: 43. A* Challenge: Synthesis of Polyesters (PET / Terylene) 43. (a) $-[O-CH_2-CH_2-O-CO-C_6H_4-CO]-_n$ (3). 43. (b) Each ester linkage formation combines $-COOH$ and $-OH$ groups, eliminating 1 molecule of H_2O per linkage (2).	[5]
Q44	Q44: 44. A* Challenge: Biodegradability of Polyesters vs Addition Polymers 44. (a) PET contains polar ester linkages ($-CO-O-$) vulnerable to nucleophilic attack and hydrolysis by environmental water/enzymes/microorganisms (3). 44. (b) Poly(ethene) consists of non-polar, unreactive strong $C-C$ and $C-H$ single bonds resistant to hydrolysis and bacterial attack (2).	[5]
Q45	Q45: 45. A* Challenge: Synthesis of Acyl Chlorides using Thionyl Chloride $SOCl_2$ 45. (a) $CH_3COOH + SOCl_2 \rightarrow CH_3COCl + SO_2(g) + HCl(g)$ (2). 45. (b) Both byproducts (SO_2 and HCl) are gases that escape automatically, leaving pure liquid acyl chloride without requiring fractional distillation (unlike liquid $POCl_3$ byproduct from PCl_5) (3).	[5]
Q46	Q46: 46. A* Challenge: Enthalpy of Combustion Trends in Monocarboxylic Acids 46. (a) $\Delta_c H$ diff: $(-874 - (-255)) = -619$; $(-1527 - (-874)) = -653$; $(-2184 - (-1527)) = -657$. Average increment ~ -650 kJ mol $^{-1}$ per $-CH_2-$ group (2). 46. (b) Each additional $-CH_2-$ group requires breaking 2 $C-H$ and 1 $C-C$ bond and forming 1 CO_2 and 1 H_2O , providing a constant increment of heat release (3).	[5]

Q47	Q47: 47. A* Challenge: Synthesis of Poly(lactic acid) PLA via Cyclic Lactide 47. (a) 6-membered ring containing 2 ester linkages: 3,6-dimethyl-1,4-dioxane-2,5-dione (3). 47. (b) Ring-opening polymerisation of dilactide with tin(II) octoate catalyst yields high molar mass biodegradable PLA polymer without releasing water (2).	[5]
Q48	Q48: 48. A* Challenge: Gas Chromatography - Mass Spectrometry (GC-MS) of Fatty Acid Methyl Esters 48. (a) High boiling point fatty acids decompose at high temperatures; methyl esters are more volatile and pass cleanly through GC column without thermal breakdown (2). 48. (b) GC separates esters by boiling point/polarity giving unique retention times (1.5); MS electron ionization gives distinct molecular ion M^+ (m/z 296 for methyl oleate vs m/z 270 for methyl palmitate) and fragmentation patterns (1.5).	[5]
Q49	Q49: 49. A* Challenge: Synthesis of Nylon-6,6 vs Terylene Comparison 49. (a) Hexane-1,6-diamine + Hexanedioic acid. Repeat unit: $-[HN-(CH_2)_6-NH-CO-(CH_2)_4-CO]-n$ (3). 49. (b) Nylon-6,6 forms strong intermolecular hydrogen bonds between N-H and C=O of adjacent chains (higher melting point $\sim 265^\circ\text{C}$); Terylene forms permanent dipole-dipole attractions between ester C=O groups (2).	[5]
Q50	Q50: 50. A* Challenge: Complete Carboxylic Acid Master Synthesis & Elucidation 50. (a) A ($C_7H_6O_2$) gives CO_2 effervescence with $NaHCO_3 \Rightarrow$ carboxylic acid $\Rightarrow$ Benzoic acid, C_6H_5COOH (1.5). B = Benzoyl chloride, C_6H_5COCl (PCl_5 test) (1.5). C = Methyl benzoate, $C_6H_5COOCH_3$ (esterification) (1.5). Nitration of benzoic acid gives 3-nitrobenzoic acid D ($-COOH$ is meta-directing) (1.5).	[6]

SECTION D: EMPIRICAL EXAMINER FAQS & COMMON TRAPS (10 FAQS) — 15C

FAQ #1: Carboxylic Acid Dimerization

Category: Intermolecular Forces

COMMON EXAMINER TRAP / MISTAKE:

Assuming carboxylic acids only form single hydrogen bonds.

MODEL ANSWER / EXAMINER EXPECTATION:

Carboxylic acids form HEAD-TO-TAIL DIMERS in liquid/gas phases held by TWO hydrogen bonds per pair. This effectively doubles their molecular mass and elevates boiling points above alcohols.

FAQ #2: NaHCO₃ Test Specificity

Category: Carbonate Effervescence

COMMON EXAMINER TRAP / MISTAKE:

Thinking phenols give a positive effervescence test with NaHCO₃.

MODEL ANSWER / EXAMINER EXPECTATION:

Phenols (pK_a ~10) are TOO WEAK to react with NaHCO₃. ONLY carboxylic acids (pK_a ~4–5) are strong enough to liberate CO₂ gas from NaHCO₃(aq). This test unequivocally distinguishes carboxylic acids from phenols.

FAQ #3: PCl₅ Reaction Conditions & Byproducts

Category: Acyl Chloride Preparation

COMMON EXAMINER TRAP / MISTAKE:

Writing water H₂O as a byproduct in PCl₅ reactions.

MODEL ANSWER / EXAMINER EXPECTATION:

PCl₅ reacts with –COOH to form acyl chloride RCOCl + POCl₃(l) + HCl(g). Reaction MUST be dry because PCl₅ and RCOCl react violently with water.

FAQ #4: Role of Concentrated H₂SO₄ in Esterification

Category: Esterification Catalyst

COMMON EXAMINER TRAP / MISTAKE:

Stating H₂SO₄ acts solely as a catalyst in esterification.

MODEL ANSWER / EXAMINER EXPECTATION:

Concentrated H₂SO₄ acts as BOTH a acid catalyst (protonates C=O) AND a DEHYDRATING AGENT (absorbs water byproduct, shifting equilibrium right to increase ester yield).

FAQ #5: Isotopic Labeling in Esterification

Category: Mechanism Proof

COMMON EXAMINER TRAP / MISTAKE:

Thinking the C–OH bond of the alcohol breaks during esterification.

MODEL ANSWER / EXAMINER EXPECTATION:

Isotopic ¹⁸O labeling proves the C–OH bond of the CARBOXYLIC ACID breaks, NOT the alcohol O–H bond. The ¹⁸O label from alcohol ends up in the ester.

FAQ #6: Inductive vs Resonance Effects on Acidity

Category: Organic Acidity

COMMON EXAMINER TRAP / MISTAKE:

Confusing electron-withdrawing (-I) with electron-donating (+I) group impacts.

MODEL ANSWER / EXAMINER EXPECTATION:

Electron-withdrawing groups (-Cl, -NO₂) pull electron density away from -COO⁻, stabilizing the anion and INCREASING acid strength. Electron-donating alkyl groups (+I) push density towards -COO⁻, destabilizing anion and DECREASING acid strength.

FAQ #7: Formic Acid (HCOOH) Unique Oxidation**Category: Formic Acid Oxidation****COMMON EXAMINER TRAP / MISTAKE:**

Treating methanoic acid as a typical unreactive carboxylic acid.

MODEL ANSWER / EXAMINER EXPECTATION:

Methanoic acid HCOOH contains an ALDEHYDE C-H proton H-C(=O)-OH. Unlike other carboxylic acids, HCOOH is EASILY OXIDIZED by Tollens', Fehling's, or KMnO₄ to CO₂ and H₂O.

FAQ #8: Soda Lime Decarboxylation Chain Length**Category: Decarboxylation****COMMON EXAMINER TRAP / MISTAKE:**

Forgetting that soda lime decarboxylation shortens the carbon chain by ONE carbon.

MODEL ANSWER / EXAMINER EXPECTATION:

Heating RCOONa with soda lime (NaOH/CaO) removes -COONa as Na₂CO₃, yielding alkane R-H with ONE LESS carbon atom than the original acid.

FAQ #9: Nitrile Hydrolysis Reaction Products**Category: Nitrile Hydrolysis****COMMON EXAMINER TRAP / MISTAKE:**

Forgetting ammonium salt byproduct in acidic nitrile hydrolysis.

MODEL ANSWER / EXAMINER EXPECTATION:

Acidic hydrolysis of nitriles yields carboxylic acid + AMMONIUM SALT: $\text{RCN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{RCOOH} + \text{NH}_4\text{Cl}$. Alkaline hydrolysis yields carboxylate salt + AMMONIA: $\text{RCN} + \text{H}_2\text{O} + \text{NaOH} \rightarrow \text{RCOONa} + \text{NH}_3$.

FAQ #10: LiAlH₄ vs NaBH₄ Reduction of Acids**Category: Reduction Selectivity****COMMON EXAMINER TRAP / MISTAKE:**

Attempting to reduce carboxylic acids using NaBH₄.

MODEL ANSWER / EXAMINER EXPECTATION:

Carboxylic acids are UNREACTIVE towards NaBH₄. You MUST use LiAlH₄ in dry ether followed by acid hydrolysis to reduce a carboxylic acid to a primary alcohol.